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UNIVERSITY OF PADUA, DEPARTMENT OF INFORMATION ENGINEERING
MASTER THESIS IN ELECTRONIC ENGINEERING

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*To my parents
and friends*



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Abstract

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Introduction

The number of wireless sensor networks (WSNs) and Internet of Things (IoT) devices has grown rapidly in recent years, raising a fundamental question: how to power all these devices without replacing batteries or requiring constant maintenance. Recent advances in low-power electronics have reduced the energy consumption of sensor nodes to the range of tens to hundreds of microwatts [3]. At such low power levels, it becomes possible to power these devices by extracting energy from the surrounding environment, removing the need for batteries and allowing the network to operate indefinitely.

Batteries are the most common power source for WSN nodes, but they have well-known limitations. In many applications, such as structural health monitoring, industrial machinery surveillance, or medical implants, replacing or recharging depleted batteries is difficult or simply not possible [1]. Using larger batteries to extend the device lifetime increases the size of the system, which is often not acceptable. In addition, the environmental cost of battery and the consequent maintenance of large-scale networks are serious obstacles to the widespread use of WSNs [2].

Energy harvesting offers a practical solution to these problems. Instead of storing a fixed amount of energy, harvesting systems collect energy from sources available in the environment such as sunlight, heat gradients, mechanical vibrations, or radio frequency signals and convert it into electrical power to supply an electronic system.

Among all available energy sources, mechanical vibrations have received the most attention in the research community, accounting for more than half of publications in this field [2]. Piezoelectric, triboelectric, and electromagnetic transducers are the most widely used mechanisms, each with different trade-offs in output voltage, power density, and complexity [3, 1]. Figure 1.1 shows

the distribution of recent publications about transducer types, highlighting the dominance of mechanical energy harvesting approaches.

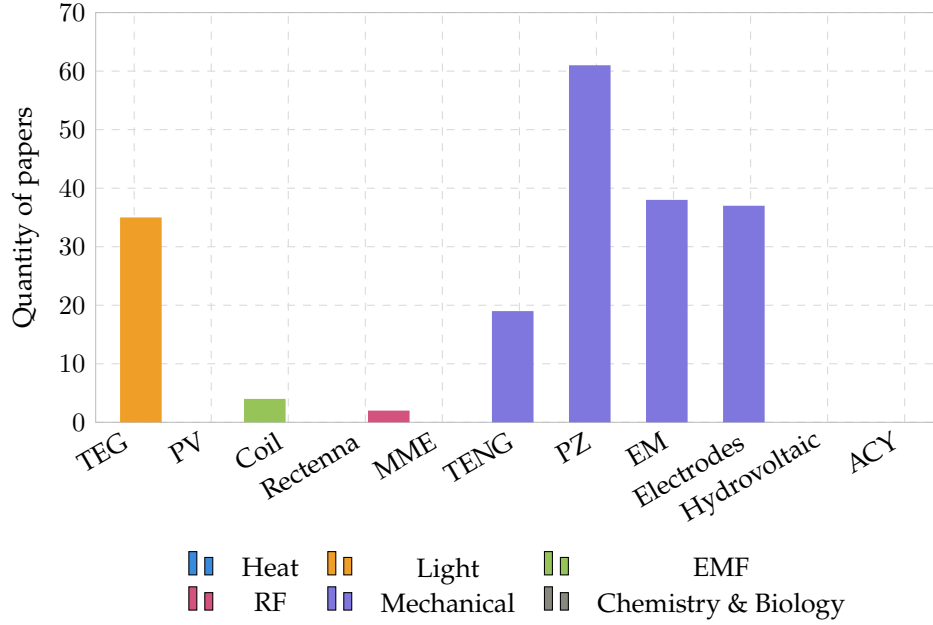


Figure 1.1: Distribution of papers based on transducer type. Adapted from [2].

Energy harvesting systems are generally divided into two categories [1]: systems that use the harvested energy immediately (*Harvest-Use*), and systems that store it first in a battery or supercapacitor before supplying it to the node (*Harvest-Store-Use*). The second approach is far more common, since most energy sources are not continuously available and some form of storage is needed to guarantee uninterrupted operation.

2

Rectifiers

sui [4]

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Acknowledgments

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